

Phys 247 : Fall 2026

Problem Set 3

Assigned 9/18, Due Monday 9/28

Submit HW by Start of Class, No Late Grades

Start early!

Problem 1 (easy)

KK 3.19 - Compound Springs

KK 3.7 - Tension, Gravity, and Friction

Problem 2 (medium)

KK 3.8 - Block and Wedge, now with Friction

Problem 3 (hard)

KK 3.12 - Sailboat Physics

Hint 1: Can you motivate $dT/d\theta = -\mu T$?

Hint 2: Carefully review example 3.7.

Problem 4 (easy)

KK 4.4 - Water Molecule

KK 4.1 - Non-uniform rod

Extra Credit (medium)

KK 4.15 - Cart Jumpers

Submission Instructions:

Scan a PDF. Either use an office machine or make clear photos with your cell phone. If you make photos, please collect images into PDF.

Upload to Gradescope on canvas. Do **not** use AI to enhance, check, or auto-latex your solutions. You're encouraged to work together in groups, but everyone must write and submit solutions individually.